

VISHESH SHEKHAWAT

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Education

M.Tech in Artificial Intelligence (Generative AI)

NIT Jalandhar

Aug 2025 – Present

CGPA: 8.78/10

B.Tech in Computer Science & Engineering

Sobhasaria Group of Institute, Sikar

2020 – 2024

CGPA: 8.45/10

Projects

Modular Adaptive Network OS (manOS)

FastAPI, PostgreSQL, FAISS, RAG

- Designed modular AI system enabling isolated learning environments with scalable orchestration and efficient resource management across instances
- Built end-to-end pipeline from ingestion and preprocessing to semantic chunking, LLM-based QA generation, and persistent storage integration
- Integrated RAG pipeline using FAISS embeddings to enable context-aware retrieval and improve answer relevance across dynamic knowledge sources
- Developed adaptive learning engine leveraging spaced repetition and performance-driven scheduling for continuous model improvement
- Implemented FastAPI backend with repository pattern ensuring clean abstraction, modularity, and efficient API orchestration

NewStack-AI – Modular LLM News Intelligence Pipeline

Python, LLMs, RAG

- Built a modular pipeline to aggregate AI-related content from YouTube transcripts and RSS feeds for structured information processing
- Transformed unstructured HTML and transcript data into clean markdown and generated LLM-based summaries for improved readability
- Implemented ranking and personalization logic to score content relevance and prevent duplicate information delivery
- Integrated retrieval-style grounding using metadata constraints to enhance factual consistency and reduce hallucinations
- Automated end-to-end pipeline for generating personalized HTML digests with centralized orchestration

Hackathons & Competitions

StatHack — Skin Disease Classification (IIT Goa)

Computer Vision

- Built CNN-based models with preprocessing and augmentation pipelines for medical image classification tasks
- Achieved **6th rank** on Kaggle evaluation leaderboard among competitive participants

Heart Disease Prediction — ML Competition

Scikit-Learn, XGBoost

- Built ensemble models on 600k+ records achieving ROC-AUC score of 0.955 through feature engineering and model tuning
- Ranked in **Top 5.4% (238 / 4370)** demonstrating strong predictive modeling performance

Achievements

- Achieved **Rank 155** on TensorTonic Platform
- Solved **300+ DSA problems** on LeetCode (Top 32%)
- Secured **Rank 1844** on GateOverflow (99.91 percentile)
- Qualified **GATE 2024 & GATE 2025**

Skills

Programming: Python (NumPy, Pandas), C/C++, SQL

Frameworks & Tools: PyTorch, Scikit-learn, FastAPI

LLM Systems & Engineering: RAG, Prompt Engineering, Pipeline Design, REST APIs